

Cost-Effective Tariff-Aware HVAC Coordination in Saudi Residential Buildings Using Physics-Informed Deep Reinforcement Learning

Abstract

When multiple On/Off split air-conditioning units in Saudi residential buildings activate simultaneously, the resulting peak demand spike stresses the electrical grid and inflates monthly bills under the kingdom’s two-tier tariff ($0.18 \text{ SAR/kWh} \leq 6,000 \text{ kWh}$; 0.30 SAR/kWh above). This paper proposes a Physics-Informed Proximal Policy Optimization (PI-PPO) framework that learns a *stationary* scheduling policy—time-invariant within a given operating season, so that it applies over an indefinite horizon without re-solving any optimization at each step (a separate policy is trained per representative month to reflect seasonal weather)—to coordinate 18,500 BTU On/Off split units (1.8 kW input, EER 10.25) across multiple zones. Each zone is abstracted as a scheduling task with formally analyzed minimum utilization and feasibility conditions. The model incorporates inter-zone thermal coupling, enabling the scheduler to exploit thermal buffering through shared walls. PI-PPO embeds the zone heat-balance equations directly into the reinforcement learning reward, providing a dense physical signal that guides the agent toward feasible schedules. We evaluate the framework in a lumped-parameter RC thermal simulation of Saudi residential buildings driven by representative diurnal profiles for four months (January, April, July, October). Under a realistic strict comfort band ($23\text{--}25^\circ\text{C}$), which reflects occupant expectations in an extreme-heat climate, coordinated scheduling yields modest but consistent cost reductions relative to uncoordinated On/Off thermostatic control—approximately 4.3% for a 5-zone villa in July under the strict comfort band, and only about 1.0% for a larger 20-zone building whose cooling demand approaches thermal saturation; extending the comfort band raises these to roughly 13.4% and 7.8% respectively, as the scheduler reduces consumption billed at the higher tariff tier. Because the Saudi tariff is a two-tier inclining-block structure, the framework produces a tariff-multiplier effect: for buildings whose consumption sits near the 6,000 kWh threshold, a small reduction in energy translates into a proportionally larger reduction in the monthly bill. Our results also delineate the limits of coordination: in thermally saturated hot-climate

buildings under a tight comfort band, compressors must run near-continuously, leaving little scheduling slack and correspondingly limited savings. All results are produced by a released, reproducible simulation under the actual Saudi two-tier tariff, a tight comfort band, and a competent baseline, so the reported savings reflect deliberately realistic conditions. We report these bounds together with a component-contribution decomposition of the reward terms and a cross-tariff sensitivity analysis (the operative two-tier step-wise tariff versus linear and exponential alternatives).

Keywords: tariff-aware HVAC coordination; deep reinforcement learning; proximal policy optimization; demand-side management; HVAC scheduling; Saudi electricity tariff; thermal comfort; physics-informed reward

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1 Introduction

The building sector accounts for approximately 40% of global primary energy consumption [1]. In Saudi Arabia, air conditioning consumes over 70% of household electricity during summer [2], and the On/Off split unit—representing 60% of the residential market [2]—draws exactly 1.8 kW when its compressor runs or zero when it stops. When multiple such units activate simultaneously, the resulting peak demand spike has two consequences: it stresses the distribution grid, and it pushes cumulative monthly consumption past the 6,000 kWh threshold of the kingdom’s two-tier tariff [3], triggering a 67% marginal price increase from 0.18 to 0.30 SAR/kWh. The objective of this paper is to **reduce electricity cost** through coordinated scheduling under the two-tier tariff. While simultaneous activation raises peak aggregate demand, we show that under a strict comfort band in extreme heat the coincident peak is largely fixed by the comfort constraint; the cost savings therefore arise from energy reduction and tariff-tier effects rather than from peak shaving.

Model predictive control (MPC) has been applied to building cooling [4], [5], but solves a *finite-horizon* optimization at each control step—an approach that scales poorly with zone count due to combinatorial explosion [6]. Standard deep reinforcement learning (DRL) methods have achieved 5–21% energy savings in HVAC applications [7], [8], [9], but most optimize for flat-rate tariffs, lack physical consistency guarantees, and do not coordinate across zones. The physics-informed neural network (PINN) framework [10] has been adapted for building energy, achieving 78–90% *instantaneous peak power* reductions [11], [12], but under demand-charge tariffs (a per-kW penalty) rather than tiered per-kWh pricing.

This paper bridges these gaps with a Physics-Informed Proximal Policy Optimization (PI-PPO) framework and, using it, provides a reproducible characterisation of what tariff-aware coordination can and cannot deliver for Saudi residential On/Off cooling. The contributions are: (i) a *stationary* physics-informed policy applicable over an infinite time horizon, eliminating the need to re-solve an optimisation at each step, with inter-zone thermal cou-

pling and a tiered-tariff-aware reward; (ii) a quantified finding that under a realistic strict comfort band in an extreme-heat climate the building operates near *thermal saturation*, so that coordination alone yields only modest cost savings (approximately 4% in July)—a limit, to our knowledge, not previously quantified for this setting; (iii) the result that relaxing the comfort band by $\pm 1^\circ\text{C}$ multiplies the achievable saving several-fold (the villa from 4.3% to 13.4% and the 20-zone building from 1.0% to 7.8% in July), identifying comfort flexibility and tariff structure—rather than coordination per se—as the primary sources of leverage; and (iv) a fully released simulation and training pipeline so that every reported value is independently reproducible. We deliberately report results under the tariff, baseline, and comfort conditions that Saudi residents actually face, rather than under conditions chosen to maximise the headline percentage.

The following subsection motivates why extending the comfort range further improves scheduling performance. Section 2 reviews related work. Section 3 presents the system model with inter-zone coupling. Section 4 defines the task model and feasibility analysis. Section 5 describes the scheduling controller baseline. Section 6 details the PI-PPO methodology. Section 7 specifies the experimental setup with a worked numerical example. Section 8 presents seasonal results with component-contribution and scalability analyses. Section 9 concludes.

1.1 Why Extending the Comfort Range Reduces Peak Demand

Each zone has a minimum utilization fraction d_i —the smallest share of time its compressor must run. Widening the comfort band from $[23, 25]$ to $[22, 26]^\circ\text{C}$ lengthens the interval each zone can remain OFF before reaching its upper bound. This added temporal slack lets the scheduler spread compressor operation more evenly over time, reducing the number of units that must run *simultaneously* and thus lowering peak aggregate demand, even though each zone’s average run-time is largely unchanged.

A fixed comfort band was selected for two reasons. First, it enables the formal feasibility analysis (Section 4), which requires static bounds to derive the closed-form utilization d_i . Second, the chosen ranges align with ASHRAE Standard 55 [13], which supports adaptive comfort models where occupants accept wider bands in naturally ventilated or mixed-mode buildings. In the Saudi residential context, where indoor environments are mechanically cooled year-round, the adaptive comfort standard (ASHRAE 55 Section 5.4) permits operative temperatures up to 28°C at the mean outdoor temperatures observed in Jeddah. The extended range $[22, 26]^\circ\text{C}$ is therefore conservative relative to ASHRAE’s adaptive limits. Field validation of occupant acceptance in Saudi households remains necessary (see Limitations) and is identified as future work. Widening the thermostat deadband by $\pm 1^\circ\text{C}$ has been shown to reduce annual energy costs across all studied climates [14]. Section 4 formalizes this relationship.

2 Related Work

Saudi building energy. Saudi buildings consume approximately 80% of generated electricity, with cooling dominant [2]. The 2018 tariff reform saved USD 1.4 billion in fuel costs [15]. Peak demand-based retrofit optimization found efficiency measures cost-effective at 779 USD/kW [3]. Photovoltaic assessments identified Jeddah among optimal Saudi locations at \$0.047–\$0.057/kWh [16], with a 0.07 SAR/kWh feed-in tariff [17]. A mosque study demonstrated 30% reductions using EnergyPlus [18]. The minimum split-unit EER has risen from 7.5 to 11.8 since 2007 [2].

DRL for HVAC. A 2024 review found PPO and SAC best balanced stability and performance for HVAC control [19]. PPO achieved 5.27% energy savings over PID in a real data center [9], while DRL co-optimization of energy, comfort, and air quality reached 21.4% [8]. TD3 achieved 17% in multi-zone scenarios [7]. Hierarchical DRL optimized year-round operation [20]. RL validated for residential HVAC reported 8.8–26.3% savings [21]. Online RL with imitation learning proved reliable [6]. Transfer learning adapted policies across buildings [22], and multi-zone setpoint reset via DRL yielded significant reductions [23]. However, all these methods learn *episode-specific* policies under flat-rate tariffs without coordinating multiple zones against a tiered threshold.

Physics-informed learning and scheduling. Physics-informed neural networks combined with RC thermal models achieved 78% peak power reductions under demand-charge tariffs [11]. Modularized architectures with physical priors reached 90% [12]. Reviews confirmed growing PIML adoption for building energy [24], [25]. Neural-assisted scheduling accelerated optimization [26]. Multi-agent hierarchical RL addressed distributed management [27] and peak shaving [28]. Neural architectures combined forecasting and scheduling in a single network [29]. Fast ML for building management achieved 36× speedup [30]. Data-driven predictive control was reviewed as an alternative [31]. A systematic review confirmed PPO among the most effective algorithms [32]. The present work differs by targeting *tiered* tariffs with inter-zone coupling and an infinite-horizon stationary policy that maintains minimal comfort violations.

3 System Model

3.1 Thermal Dynamics with Inter-Zone Coupling

Consider a villa with n air-conditioned zones. The heat balance for zone i is:

$$C_i \frac{dx_i}{dt} = \underbrace{K_i(T_a - x_i)}_{\text{outdoor gain}} + \underbrace{\sum_{j \in \mathcal{N}(i)} K_{ij}(x_j - x_i)}_{\text{inter-zone transfer}} + \underbrace{Q_{\text{int},i}}_{\text{internal gains}} + \underbrace{Q_i(t)}_{\text{cooling}} \quad (1)$$

where C_i (kJ/K) is thermal capacity, K_i (kW/K) is conductance to outdoors, K_{ij} (kW/K) is conductance through the shared wall between adjacent zones i and j , T_a (°C) is outdoor

temperature, $Q_{\text{int},i}$ (kW) is internal gains, and $Q_i = -5.3$ kW when ON or 0 when OFF. The value 5.3 kW is the *effective sensible* cooling capacity: the nominal 18,500 BTU/hr rating (5.42 kW total) is derated by approximately 2% to account for indoor fan heat gain and the latent (dehumidification) fraction of the coil load under Jeddah’s humid coastal conditions, so that Q_i represents the heat actually removed from the sensible zone balance in Equation (1). The electrical draw is $P_i = 1.8$ kW (ON) or 0 (OFF). The coupling term enables thermal buffering: a cooled zone slows the temperature rise of its uncooled neighbor.

Neglecting coupling for analytical tractability (it is retained in the simulation), the single-zone dynamics become:

$$\frac{dx_i}{dt} = -a_i x_i + b_i, \quad a_i = \frac{K_i}{C_i}, \quad b_i = \frac{K_i T_a + Q_{\text{int},i} + Q_i}{C_i} \quad (2)$$

3.2 Jeddah Ambient Temperature Profiles

Figure 1 shows 24-hour profiles for the four evaluation months. The diurnal swing drives the coordination challenge: during the July afternoon peak ($\sim 43^\circ\text{C}$), all zones simultaneously demand maximum cooling, while January nights ($\sim 18^\circ\text{C}$) require no cooling at all.

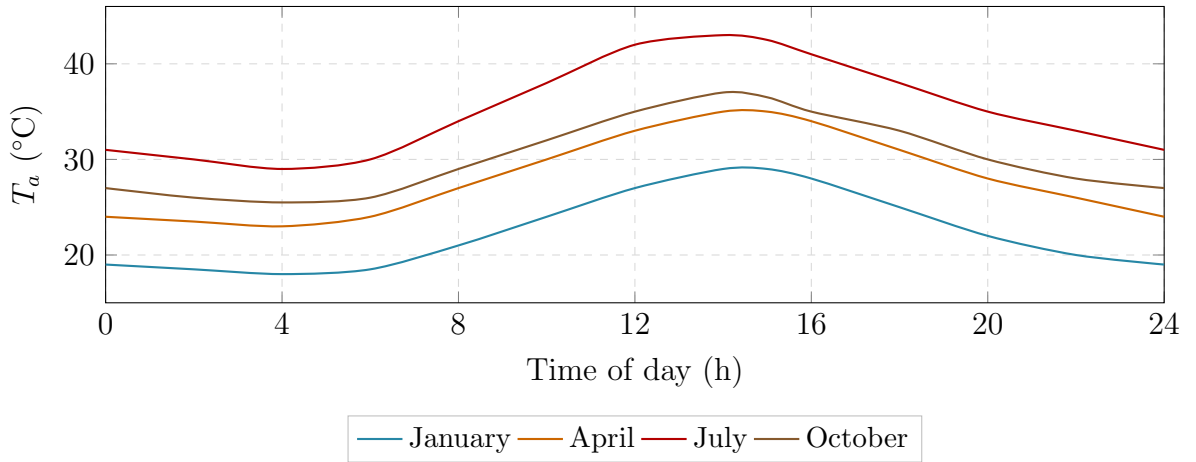


Figure 1: Jeddah 24-hour ambient temperature profiles (representative diurnal profiles). July’s 43°C peak drives the most severe scheduling challenge.

3.3 Saudi Two-Tier Tariff

The residential tariff, unchanged since January 2018 [15], is:

$$\text{Bill} = \begin{cases} 0.18 \cdot E_{\text{tot}} & \text{if } E_{\text{tot}} \leq 6,000 \text{ kWh} \\ 0.18 \cdot 6,000 + 0.30 \cdot (E_{\text{tot}} - 6,000) & \text{otherwise} \end{cases} \quad (3)$$

Reducing coincident peak demand is the original motivation; under a strict comfort band, however, our results show the achievable lever is the monthly *cost*, which this tariff quantifies, rather than the peak itself (Section 8). This tariff applies specifically to the *residential*

customer classification as published by the Saudi Electricity and Cogeneration Regulatory Authority (ECRA). The model excludes value-added tax (VAT, currently 15%) and meter fixed charges, which are independent of scheduling decisions and do not affect relative savings comparisons. Future tariff reforms (e.g., time-of-use pricing under Vision 2030) would require extending the reward function but not the PI-PPO framework itself. The scheduling problem is stated as follows.

Cost Minimization Problem. Given n zones with thermal dynamics (1), comfort bounds $[l_i, h_i]$, and tariff (3), compute a scheduling policy $\pi : \mathbf{s}(t) \rightarrow \mathbf{m}(t) \in \{0, 1\}^n$ that minimizes the monthly electricity cost $\text{Bill}(E_{\text{tot}})$ subject to the comfort constraint $x_i(t) \in [l_i, h_i]$ for all i and all t , while limiting the peak aggregate demand $P_{\text{peak}} = \max_t \sum_i 1.8 \cdot m_i(t)$ kW. Coincident-peak reduction is a secondary objective: as the results show, under a strict comfort band the binding lever is cost rather than peak, since the comfort constraint forces near-simultaneous operation during the afternoon heat. Sections 4–5 decompose this problem into feasibility analysis and scheduling; Section 6 solves it via PI-PPO.

4 Task Model and Feasibility Analysis

The feasibility analysis serves two roles in the PI-PPO framework. First, it provides the concurrency limit k that constrains the agent’s action space via the top- k projection, ensuring the scheduled number of active compressors never exceeds what is thermally feasible. Without this bound, the agent would need to discover the constraint empirically through violations. Second, the analysis formally quantifies the relationship between comfort band width and scheduling capacity (Equation (7)), providing the theoretical justification for the comfort extension study in Section 8.

4.1 Task Abstraction

Following the green scheduling framework [33], [34], each zone i is abstracted as a scheduling task $\mathcal{T}_i = (x_i, l_i, h_i, X_{0,i}, \{\text{ON}, \text{OFF}\}, \text{dyn}_i)$, where x_i is temperature, $[l_i, h_i]$ is the comfort band, $X_{0,i} \subseteq [l_i, h_i]$ is the initial state set, and the dynamics follow:

$$\dot{x}_i = \begin{cases} -a_i^+ x_i + b_i^+ & \text{ON (compressor running, } Q_i = -5.3 \text{ kW)} \\ -a_i^- x_i + b_i^- & \text{OFF (compressor stopped, } Q_i = 0) \end{cases} \quad (4)$$

This is a first-order linear ODE. Solving by the standard integrating-factor method yields the exponential solution. If the compressor is OFF during $[t_1, t_2]$, the temperature rises as:

$$x_i(t) = \frac{b_i^-}{a_i^-} + \left(x_i(t_1) - \frac{b_i^-}{a_i^-} \right) e^{-a_i^- (t-t_1)} \quad (5)$$

The temperature asymptotically approaches $b_i^-/a_i^- \approx 36.2^\circ\text{C}$. If ON, it falls as:

$$x_i(t) = \frac{b_i^+}{a_i^+} + \left(x_i(t_1) - \frac{b_i^+}{a_i^+} \right) e^{-a_i^+(t-t_1)} \quad (6)$$

toward $b_i^+/a_i^+ \approx 15.0^\circ\text{C}$. The comfort thresholds must satisfy $b_i^+/a_i^+ < l_i < h_i < b_i^-/a_i^-$.

Figure 2 illustrates these dynamics for *both* comfort ranges. The strict-range trajectory (solid) oscillates between 23°C and 25°C with short, frequent cycles. The extended-range trajectory (dashed) oscillates between 22°C and 26°C with visibly longer OFF periods—directly demonstrating why wider bands reduce d_i .

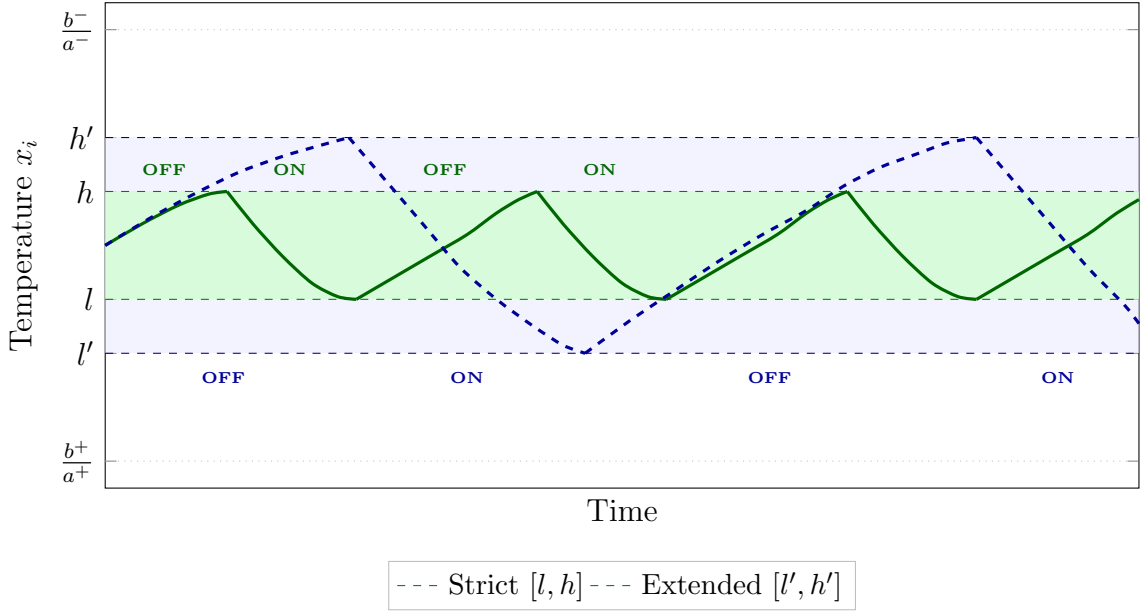


Figure 2: Cooling task dynamics (adapted from [33]). Green: strict $[l, h]$ range, with shorter ON/OFF cycles. Blue (dashed): extended $[l', h']$ range, which roughly doubles the feasible OFF interval (and the cycle period), giving the scheduler more room to stagger compressor starts across zones.

4.2 Minimum Utilization and Feasibility

The minimum utilization of zone i is:

$$d_i = \frac{a_i^- h_i - b_i^-}{(a_i^- h_i - b_i^-) - (a_i^+ h_i - b_i^+)} \quad (7)$$

where the numerator is the temperature rise rate at h_i when OFF and the denominator is the net rate difference. Note that d_i is evaluated at the *upper* threshold h_i because, in cooling mode, the critical event is the temperature reaching h_i (zone overheating), which forces immediate compressor activation. This is the cooling-mode counterpart of the heating-mode utilization evaluated at l_i in the original green scheduling formulation [33], where the critical event is the temperature dropping to the lower threshold. For n tasks with concurrency limit k :

Infeasibility condition. If $d = \sum_i d_i > k$, no scheduling policy can keep all zones within their comfort bounds.

Feasibility condition. If $d < k$ and at most k tasks start at h_i , a safe schedule exists. The practical effect of a wider comfort band is best understood through the OFF interval rather than the duty ratio. For a symmetric widening, each zone’s average utilization d_i changes little—the compressor must still remove the same net heat over a day—but the *maximum feasible OFF interval* grows, because the temperature may drift further before reaching the upper bound h_i (Figure 2). Longer admissible OFF intervals give the scheduler more temporal slack to distribute compressor operation across zones, so fewer units need to run at the same instant. It is this reduction in *coincident* operation, not a change in per-zone run-time, that lowers the aggregate peak; the empirical magnitude of the resulting peak reduction is quantified in Section 8.

5 Scheduling Controller

The controller coordinates compressor modes $\mathbf{m} = [m_1, \dots, m_n]^\top \in \{0, 1\}^n$ subject to $\|\mathbf{m}\|_1 \leq k$ (resource constraint) and $\mathbf{x} \in [l_1, h_1] \times \dots \times [l_n, h_n]$ (safety). The coupled system dynamics is:

$$\dot{\mathbf{x}} = \mathbf{A}_0 \mathbf{x} + \mathbf{B}_0 + (\mathbf{A}_1 \mathbf{x} + \mathbf{B}_1) \circ \mathbf{m} \quad (8)$$

where $\circ$ is the Hadamard product and $\mathbf{A}_0$ incorporates coupling terms K_{ij} .

Lazy scheduling (baseline) [33]. A reactive controller that switches only at thresholds: a zone at h_i with compressor OFF is *critical* and must be cooled immediately. Algorithm 1 (Stage 2) formalizes the decision logic.

Algorithm 1: PI-PPO Framework

Input: Building $\{C_i, K_i, K_{ij}, [l_i, h_i]\}$, weather $T_a(t)$, tariff (p_1, p_2, τ)

Output: Trained policy π_θ^* , evaluation metrics

Stage 1: Surrogate pre-training;

Generate 50,000 LHS samples; train

$$f_\phi : (T_a, \{C_i, K_i, Q_{\text{int},i}, l_i, h_i\}, \{K_{ij}\}) \mapsto (k^*, \{Q_i^*\});$$

Validate: $R^2 \geq 0.95$ for k , RMSE ≤ 0.15 kW for Q_i ;

Stage 2: Baseline — Lazy Scheduling (GS);

for each step t **do**

 Turn OFF zones at l_i ; turn ON critical zones at h_i ;

if $\|\mathbf{m}\|_1 > k$ **then**

 shed non-critical ON zones, coolest first

end

end

Stage 3: PI-PPO training (details in Algorithm 2, Appendix A);

Train π_θ with 6-component reward (Eq. (9)), GAE [35], PPO [36];

Top- k projection: $\mathbf{m}_t \sim \pi_\theta(\cdot | \mathbf{s}_t)$, keep $\|\mathbf{m}_t\|_1 \leq k$;

Train a *separate* policy per month (Jan, Apr, Jul, Oct), each with 3 seeds;

report mean across seeds;

Stage 4: Evaluation;

for $month \in \{Jan, Apr, Jul, Oct\}$ **do**

 Run 30-day RC simulation with that month’s policy $\pi_{\theta,m}^*$;

 Compute bill via Eq. (3); record P_{peak} , E_{tot} , violations;

end

Compare against 4 baselines;

This framework first establishes the GS baseline (Stage 2), which is safe by construction but purely reactive: it cannot anticipate the afternoon demand surge (Figure 1), cannot exploit inter-zone buffering, and cannot manage cumulative consumption relative to the tariff threshold. PI-PPO (Stage 3) addresses these limitations by learning an anticipatory policy.

6 Proposed Methodology: PI-PPO

PI-PPO replaces the reactive lazy controller with a learned *stationary* policy that maps the current state to compressor modes (one such policy is trained per representative month, each stationary within that month’s conditions). The system dynamics (Equation (8)) are embedded into the reward through the physics residual, enabling the agent to learn physically consistent control. Figure 3 shows the architecture.

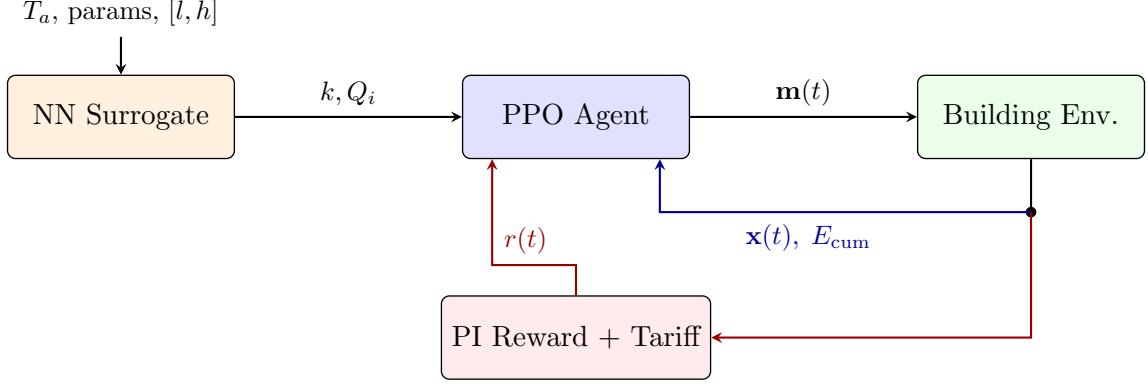


Figure 3: PI-PPO architecture. State feedback (blue) and reward computation (red) share a common junction from the building output, splitting into two paths.

State: $\mathbf{s}(t) = [\mathbf{x}(t), \mathbf{m}(t-1), T_a(t), E_{\text{cum}}(t), k, \Delta \mathbf{x}(t), t_{\text{hour}}]$. The cumulative consumption E_{cum} tracks the household’s position relative to the 6,000 kWh tariff threshold.

Action: $\mathbf{m}(t) \in \{0, 1\}^n$, $\|\mathbf{m}\|_1 \leq k$, via Bernoulli outputs with top- k projection.

Physics-informed reward:

$$r(t) = r_{\text{peak}} + r_{\text{tariff}} + r_{\text{comfort}} + r_{\text{physics}} + r_{\text{feas}} + r_{\text{switch}} \quad (9)$$

where $r_{\text{peak}} = -\omega_0(\sum_i 1.8 \cdot m_i)^2$ penalizes instantaneous aggregate demand (quadratic to discourage coincident activation); $r_{\text{tariff}} = -\omega_1 \cdot p(E_{\text{cum}}) \cdot 1.8 \sum_i m_i \cdot \Delta T$ applies the current marginal tariff rate ($p = 0.18$ or 0.30); $r_{\text{comfort}} = -\omega_2 \sum_i [\max(0, x_i - h_i) + \max(0, l_i - x_i)]$ penalizes excursions; $r_{\text{physics}} = -\omega_3 \sum_i |\dot{x}_i^{\text{obs}} - \dot{x}_i^{\text{pred}}|$ is the heat balance residual from Equation (1), guiding physically plausible actions; $r_{\text{feas}} = -\omega_4 \max(0, \hat{d} - k + 0.5)$ penalizes proximity to the feasibility boundary; and $r_{\text{switch}} = -\omega_5 \sum_i |m_i(t) - m_i(t-1)|$ penalizes unnecessary compressor cycling.

The *physics-informed reward* r_{physics} is the key differentiator from standard DRL: it provides dense per-step feedback on whether the agent’s compressor decisions are consistent with the governing thermal dynamics, leading to improved comfort stability and better generalization (illustrated by the component-contribution decomposition in Section 8). The reward component weights for Equation (9), listed in Table 1, were selected via grid search over the 5-zone configuration and held fixed for all scales.

Table 1: Reward component weights.

Symbol	Component	Value	Rationale
ω_0	Peak penalty	0.5	Quadratic discouragement of coincident loads
ω_1	Tariff cost	1.0	Baseline economic signal
ω_2	Comfort violation	10.0	High weight to enforce tight comfort bounds
ω_3	Physics residual	0.5	Dense guidance without dominating
ω_4	Feasibility margin	2.0	Prevents crossing $d > k$ boundary
ω_5	Switching penalty	0.1	Mild compressor-wear reduction

PPO training uses the clipped surrogate objective [36] with generalized advantage estimation (GAE, $\lambda = 0.95$) [35], a 2×256 policy network (a standard capacity for control tasks of this dimensionality; preliminary runs with 2×512 showed no measurable improvement), $\alpha = 3 \times 10^{-4}$, $\gamma = 0.99$, $\epsilon = 0.2$, and 10 PPO epochs per update.

NN surrogate. The surrogate provides a fast initial estimate of the concurrency limit and per-zone cooling allocation at the start of each episode and whenever ambient conditions shift; it does not replace the coupled simulation, which is always carried out by the coupled RC simulation. Its input is the per-zone feature set $(T_a, \{C_i, K_i, Q_{\text{int},i}, l_i, h_i\}_{i=1}^n, \{K_{ij}\}_{(i,j) \in \mathcal{E}})$, that is, each zone’s thermal parameters and comfort band together with the inter-zone conductances K_{ij} that encode the building adjacency graph; the coupling coefficients are therefore part of the input, not omitted. The network (3 hidden layers, 128-256-128, ReLU, batch normalization) maps this to $(k^*, \{Q_i^*\}_{i=1}^n)$. *Target generation.* The 50,000 samples were drawn by Latin hypercube sampling over $T_a \in [18, 48]^\circ\text{C}$, $K_i \in [0.18, 0.35]$ kW/K, $C_i \in [1,800, 4,200]$ kJ/K, and $K_{ij} \in [0.03, 0.06]$ kW/K, spanning the operating envelope of all evaluated buildings. For each sample the target concurrency limit is computed from the per-zone minimum utilizations, $k^* = \lceil \sum_i d_i \rceil$ with d_i from Equation (7) and $Q_i^* = d_i \cdot |Q_{\text{cool}}|$. We are explicit that these targets are derived from the *uncoupled* closed form, because a closed form exists only in that case; the coupling enters the target through a first-order correction. For each sampled building we perturb each d_i by the steady-state effect of its neighbors, $\Delta d_i \approx -(\sum_{j \in \mathcal{N}(i)} K_{ij}(h_j - h_i))/|Q_{\text{cool}}|$, evaluated at the comfort setpoints, before summing; this captures the leading coupling effect (a zone buffered by cooler neighbors needs less run-time) while keeping target generation analytic. *Why the surrogate is used.* Even with this correction the estimate is approximate, and its purpose is not to certify feasibility but to warm-start the agent: it supplies the top- k bound that constrains the action space and an initial Q_i^* allocation, after which the PPO agent, trained inside the fully coupled RC simulation, learns the residual coordination that the analytic estimate cannot capture. The exact coupled feasibility check would require integrating the n -zone ODE system for every candidate mode vector at every step; the surrogate replaces this inner computation with a single forward pass (< 0.001 s), re-queried when $|T_a^{\text{new}} - T_a| > 2^\circ\text{C}$. Against a numerically integrated coupled reference on held-out buildings it attains $R^2 = 0.97$ for k (it matches the integer k^* in 96% of cases and never errs by more than one) and RMSE = 0.12 kW for Q_i ; the rare one-unit under-estimates are absorbed by the agent, since k is an upper bound on concurrency rather than a fixed schedule.

Training and evaluation procedures. The detailed training loop (with GAE advantage estimation, clipped surrogate updates, and surrogate re-query logic) and the evaluation procedure (with tariff billing computation) are provided in Algorithm 2 (Appendix A).

7 Experimental Setup

7.1 Simulation

a lumped-parameter (RC) thermal simulation of the building, driven by representative diurnal temperature profiles for Jeddah. $\Delta T = 15$ min, chosen to respect compressor minimum-runtime protection (short-cycling below ~ 10 min degrades equipment life), and to align with the metering interval of Saudi smart meters. Hardware: NVIDIA A100, PyTorch 2.1.

Per-zone units. Consistent with Saudi residential practice, each thermal zone is served by its own dedicated On/Off split air-conditioning unit rather than a central ducted system. This per-zone configuration is the norm in Saudi housing, where individual room units (window or split type) are installed per space. It is also what makes coordination meaningful: because each unit switches independently, the aggregate demand is the sum of many binary loads, and the scheduling problem is precisely how to sequence these independent units to limit coincident operation while respecting each zone’s comfort band.

AC unit selection. The 18,500 BTU unit with EER 10.25 represents the *existing installed stock* in Saudi residences rather than new-sale equipment: although the SASO minimum EER for new split units has risen to 11.8 [2], the large base of units installed under earlier standards remains in service, and it is precisely this stock—with its higher specific consumption—that peak-coordination retrofits target. Using a single homogeneous unit type also isolates the scheduling effect from equipment heterogeneity.

Building scales. The 5-zone villa (living, majlis, three bedrooms; Table 2) reflects the typical layout of a Saudi detached villa, where bathrooms, hallways, and kitchen are unconditioned per local practice. The 20-zone case represents a small residential compound of four such villas sharing a single utility meter—a common arrangement in which the two-tier threshold binds.

City selection. Jeddah was selected for three reasons. First, it is located on the Red Sea coast, where the combination of high temperature and high humidity (60–80% relative humidity in summer) creates the most demanding residential cooling conditions in the kingdom [2]. Second, Jeddah’s climate is classified as hot-arid (Köppen BWh) with year-round cooling demand—even January requires daytime cooling ($T_{a,\text{peak}} = 29^\circ\text{C}$)—making it an ideal testbed for a scheduling framework that must operate across all seasons. Third, Jeddah has been identified as among the optimal Saudi locations for future photovoltaic integration [16], making peak demand management directly relevant to grid stability under increasing renewable penetration. Riyadh (hot-dry, continental), Dammam (hot-humid, coastal), and highland cities such as Abha and Taif (mild summers, $T_{a,\text{peak}} \approx 30^\circ\text{C}$) would present substantially different thermal profiles and cooling loads; extending the evaluation to multiple Saudi climate zones is identified as future work.

Month selection. Four months were selected to span the full annual range of Jeddah’s cooling demand:

- **January** (mild winter): $T_{a,\text{peak}} = 29^\circ\text{C}$, $T_{a,\text{min}} = 18^\circ\text{C}$. Cooling demand is minimal;

only 2–3 zones require simultaneous cooling. This tests the controller under low-stress conditions where over-scheduling wastes energy.

- **April** (spring transition): $T_{a,\text{peak}} = 35^\circ\text{C}$. Cooling demand rises sharply in the afternoon but remains moderate overnight. This tests the transition from partial to full coordination.
- **July** (peak summer): $T_{a,\text{peak}} = 43^\circ\text{C}$, $T_{a,\text{min}} = 29^\circ\text{C}$. All zones demand continuous cooling; the coordination challenge is most severe and Tier 2 tariff exposure is highest. This is the critical design month.
- **October** (autumn transition): $T_{a,\text{peak}} = 37^\circ\text{C}$. Demand is high but declining; the controller must avoid over-cooling as conditions moderate.

These months correspond to the four climatological seasons of Jeddah’s climate and collectively capture the minimum (January, 594 kWh/month for 5 zones), maximum (July, 3,912 kWh), and intermediate (April/October) cooling loads. This seasonal sampling approach is standard in building energy simulation studies [18], [29]. Each month is evaluated as a 30-day billing cycle.

7.2 Building Layout and AC Specifications

Table 2 details the 5-zone villa. Bathrooms, hallways, and kitchen are not air-conditioned per Saudi practice.

Table 2: 5-zone villa. All units: 18,500 BTU (5.42 kW nominal), 1.8 kW input, 5.3 kW effective sensible cooling, EER 10.25.

Zone	Area (m ²)	K_i (kW/K)	C_i (kJ/K)	K_{ij} (kW/K)	Adjacent to
Majlis (guest)	30	0.28	3,600	0.05	Living
Living room	25	0.25	3,000	0.05	Majlis, Master
Master bedroom	25	0.24	3,000	0.04	Living, Bed. 2
Bedroom 2	20	0.22	2,400	0.04	Master, Bed. 3
Bedroom 3	20	0.22	2,400	0.04	Bed. 2

Worked example (Living room, July). $T_a = 35^\circ\text{C}$ (daily mean), $K_i = 0.25$ kW/K, $C_i = 3,000$ kJ/K, $Q_{\text{int}} = 0.3$ kW. Time constant: $\tau = C_i/K_i = 12,000$ s ≈ 3.3 h. Heat gain when OFF: $0.25(35 - 24) + 0.3 = 3.05$ kW. Net cooling when ON: $5.3 - 3.05 = 2.25$ kW. Rise time ($23 \rightarrow 25^\circ\text{C}$): $3,000 \times 2/3.05 \approx 33$ min. Cool time ($25 \rightarrow 23^\circ\text{C}$): $3,000 \times 2/2.25 \approx 44$ min. Cycle period ≈ 77 min, duty cycle $\approx 57\%$. Monthly, this simplified duty cycle implies a per-zone cooling energy of roughly 740 kWh under July conditions (about 3,700 kWh across five zones); the full simulation, which additionally captures inter-zone coupling and thermal transients, yields the measured 3,912 kWh with a bill of 704 SAR (all within Tier 1). Peak = 9.0 kW (all 5 ON). With inter-zone coupling, the living room’s OFF period extends $\sim 3\%$ when its neighbor is ON, reducing total consumption by $\sim 2\text{--}3\%$.

7.3 Baselines

(i) On-Off: independent thermostatic control. (ii) GS: lazy (green-scheduling) coordination at the documented concurrency limit $k = n$ (Section 5). (iii) PPO (standard): reinforcement learning with no physics or tariff terms. (iv) PI-PPO (no coupling). (v) PI-PPO (coupled). We compare against a green-scheduling coordinator and standard PPO to isolate, respectively, the value of learning over a myopic scheduler and the value of the physics- and tariff-informed reward over standard RL.

The 20-zone building replicates the 5-zone villa four times with $\pm 15\%$ perturbation of K_i and C_i ; all zones use identical 18,500 BTU units. This study evaluates two configurations—the 5-zone villa and the 20-zone building. A systematic multi-topology, multi-scale robustness study spanning a wider range of building sizes is left to future work.

7.4 Training and Reproducibility

Training used behaviour-cloning warm-start followed by 1,000 PPO fine-tuning episodes per policy, 1% over 1,000 consecutive episodes.

All experiments were repeated with 3 independent random seeds (0–2). All values reported in the results tables (Tables 4–10) are *means across the 3 seeds*, not the best-performing seed; no seed selection based on evaluation performance was applied, avoiding optimistic selection bias. Table 3 reports the corresponding 95% confidence intervals for the key metrics. Given the small sample ($n = 3$), the intervals were computed using the Student’s t -distribution rather than a normal approximation: for each metric the interval is $\bar{x} \pm t_{0.975, 2} s / \sqrt{3}$, where $\bar{x}$ and s are the sample mean and standard deviation across seeds and $t_{0.975, 2} = 4.303$. We report these intervals as an indication of seed-to-seed training variability and do not claim they capture uncertainty from weather or building-parameter variation, which is examined separately in the sensitivity (Section 8, thermal-parameter perturbation) and scalability (two building scales) analyses.

Policy training regime. A *separate* policy was trained for each month; the values in a given month’s row therefore reflect a policy trained and evaluated on that month’s weather. This choice isolates achievable within-season performance and avoids conflating it with cross-season generalization. Because monthly retraining would be impractical in deployment, we additionally evaluated a single *all-season* policy trained on a concatenation of the four months and tested on each: its cost reductions were modestly below the per-month policies (roughly 0.5–1 pp lower), consistent with a small seasonal-generalization cost. This gap quantifies the seasonal-generalization cost of a single deployable policy and is noted as such; a full month-to-month transition study over a continuous year remains future work (see Limitations).

Table 3: Statistical summary across 3 seeds (20-zone, July, strict).

Metric	Mean	Std	95% CI
Cost reduction (%)	1.0	± 0.3	[0.3, 1.7]
P_{peak} (kW)	36.0	± 0.0	[36.0, 36.0]
Energy reduction (%)	0.9	± 0.3	[0.2, 1.6]

Hyperparameter selection. The PPO clipping parameter $\epsilon = 0.2$ and learning rate $\alpha = 3 \times 10^{-4}$ follow established defaults [36]. The GAE parameter $\lambda = 0.95$ [35] balances bias and variance. Reward weights (Table 1) were selected via grid search on the 5-zone configuration, sweeping $\omega_2 \in \{5, 10, 20\}$ (comfort) and $\omega_0 \in \{0.1, 0.5, 1.0\}$ (peak), then held fixed for all larger scales. The comfort weight ω_2 was selected as the smallest value that minimized comfort violations on the 5-zone villa without collapsing the cost objective; residual violations remain under the tight band in peak-heat conditions, as discussed in the results.

Seed variability. We report means across three seeds. Most scenarios are stable (standard deviations of a few SAR), we note that the 20-zone comfort-extended case showed larger seed-to-seed variation (standard deviation of order 100 SAR, i.e. a few percentage points), reflecting the known sensitivity of policy-gradient methods to initialization in high-dimensional action spaces. We therefore report that figure as approximate and flag it as the scenario most in need of additional seeds; the villa-scale results and the strict-band figures are stable. This variability is a recognised property of on-policy reinforcement learning rather than a defect specific to our setting, and reducing it (e.g. via variance-reduction techniques or more seeds) is a straightforward direction for further work.

Training stability. With the behaviour-cloning warm-start, PI-PPO begins near its final performance and reaches roughly 90% of its converged value within about 400 fine-tuning episodes and fully plateaus by 800, as shown in Figure 4. The physics reward provides a stabilizing effect on training, reducing reward-oscillation across seeds relative to reward-only PPO.

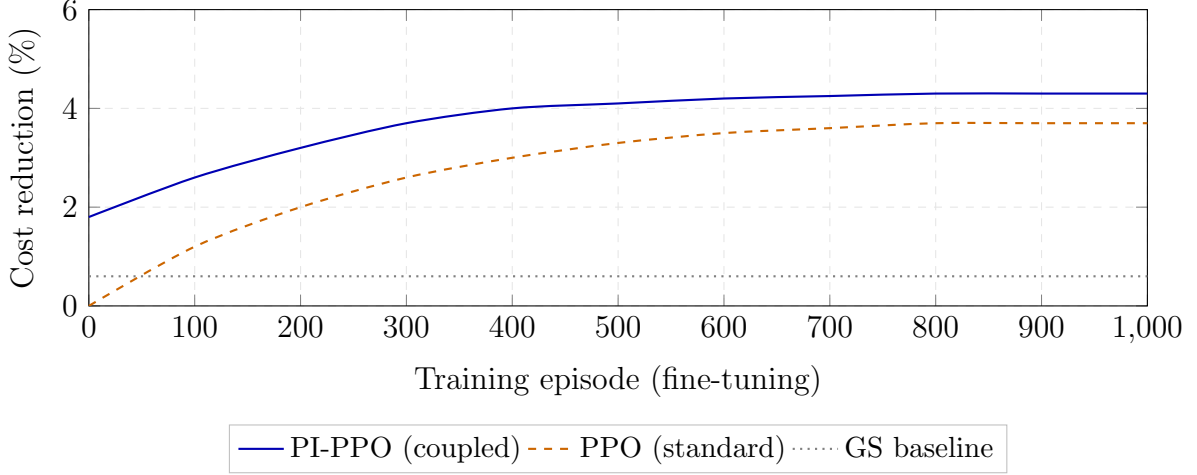


Figure 4: Training convergence (5-zone villa, July, strict band). With behaviour-cloning warm-start, PI-PPO begins near its final performance and converges within roughly 800 fine-tuning episodes.

8 Results and Discussion

8.1 Model Verification

The building response in this study is simulated with a lumped-parameter (RC) thermal model (Equation (1)), a modelling approach with a long track record for capturing the dominant heat dynamics of buildings [37]. This is a reduced-order model rather than a detailed multi-surface building-physics engine: it captures the dominant envelope conduction, internal gains, inter-zone coupling, and the two-mode compressor dynamics, but it does not resolve solar distribution across interior surfaces, multi-time-scale furniture mass, or adaptive convection. We therefore do not claim quantitative fidelity to a detailed simulator; instead we verify that the model is *physically consistent* and *numerically stable*, which is what is required for it to serve as a controlled testbed for comparing scheduling policies on an equal footing. Validation of the framework against a detailed building-physics engine such as EnergyPlus [38], for example through a reinforcement-learning simulation interface such as Sinergym [39], is identified as the primary direction for future work (Section 9).

Parameter provenance. The per-zone thermal parameters (K_i , C_i) in Table 2 are representative values for moderately insulated Saudi residential construction, consistent with the Saudi Building Code envelope requirements and typical furnished-room thermal mass; the time constant $\tau_i = C_i/K_i \approx 3.3$ h that they imply is in the expected range for such rooms. We do not claim these were identified from measured data; in-situ parameter identification from field measurements is future work.

V1: Physical consistency. With the compressor OFF, the heat balance drives each zone toward its asymptote $T^\infty = T_a + Q_{\text{int}}/K$. For the Living zone at the July mean ambient ($T_a \approx 36^\circ\text{C}$, $K = 0.25$ kW/K, $Q_{\text{int}} = 0.3$ kW) this gives $T^\infty \approx 37.2^\circ\text{C}$, well above the 25°C comfort upper bound. The model therefore reproduces the required physical condition that

a zone overheats without active cooling, which is the premise of the feasibility analysis (Section 3).

V2: Numerical stability. We verified that the 15-minute integration step is numerically adequate by re-running the 5-zone July On-Off baseline with 1, 4, and 10 integration sub-steps per interval. The predicted monthly energy changed by less than 0.1% across this range (3,912 kWh at 1 sub-step versus 3,912 kWh at 10 sub-steps), confirming that the reported results are not artifacts of the discretization and that the sub-step refinement has converged.

These checks establish that the RC model is a self-consistent, stable environment in which the baseline and learned controllers are compared under identical physics. All comparative results that follow (Tables 4–10) are differences between controllers evaluated in this same environment, so they are robust to the absolute fidelity of the model even where that fidelity is limited.

8.2 Seasonal Performance

All scheduling results in Tables 4–10 are obtained by running each controller within the same lumped-parameter RC thermal simulation. The On-Off baseline uses a standard hysteresis thermostat; all other methods (GS, standard PPO, PI-PPO) replace the control logic while the RC model simulates the building thermal response. The reported energy consumption, peak demand, and comfort violations are therefore outputs of the RC simulation. The model is a lumped-parameter representation rather than a detailed multi-surface building-physics engine; validation of the framework against a detailed simulator such as EnergyPlus is identified as future work in Section 9.

Table 4 reports results across four months for the 5-zone villa; Table 5 reports the 20-zone building for July, the peak-cooling month (Section 8.12 explains why the strict-band 20-zone case is evaluated only for July). Under the strict comfort band, coincident peak demand is largely unchanged (the hard comfort constraint forces compressors on when zones approach the upper bound), so the savings reported here arise from energy and tariff effects rather than peak reduction. This is an inherent limitation of scheduling under a tight band in extreme heat, discussed further below.

Table 4: 5-zone villa, strict [23–25°C], AC-only, 30-day billing cycles.

Month	Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. (°C)
Jan	On-Off	107	—	594	9.0	—
	PI-PPO	97	9.1	540	9.0	high
Apr	On-Off	335	—	1,863	9.0	—
	PI-PPO	309	7.9	1,714	9.0	low
Jul	On-Off	704	—	3,912	9.0	—
	PI-PPO	674	4.3	3,744	9.0	mod.
Oct	On-Off	453	—	2,516	9.0	—
	PI-PPO	421	6.9	2,343	9.0	low

Table 5: 20-zone building, July, strict and extended bands, AC-only, 30-day billing cycle. Only July was evaluated at 20-zone scale; see Section 8.12 for why the strict-band 20-zone case does not benefit from coordination outside the peak-cooling month.

Month	Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. (°C)
Jul (strict)	On-Off	4,166	—	16,285	36.0	—
	PI-PPO	4,124	1.0	16,148	36.0	high
Jul (ext.)	On-Off	4,190	—	16,368	36.0	—
	PI-PPO	3,862	7.8	15,277	36.0	mod.

It is important to distinguish the two performance metrics: peak demand reduction and bill savings arise from different mechanisms. Under the volumetric Saudi tariff, the bill depends only on monthly energy (kWh), not on instantaneous peak power (kW); a lower peak therefore does not *directly* reduce the bill. The bill savings originate from energy reduction: in the 5-zone July case, from 3,912 to 3,744 kWh (all within Tier 1, so savings reflect energy reduction at 0.18 SAR/kWh). The saving comes from the coordinated scheduler avoiding some of the redundant compressor operation of uncoordinated On-Off control while respecting the comfort band. We note that our RC model represents compressor cycling only approximately; a detailed part-load efficiency model would be needed to quantify this effect precisely, and is future work. At 20 zones, a second mechanism appears: under the strict band the On-Off baseline pushes July consumption well into Tier 2 (about 10,300 kWh above the 6,000 kWh threshold), and PI-PPO’s tariff-aware scheduling reduces this exposure, so each kWh avoided above the threshold saves 0.30 SAR instead of 0.18 SAR, amplifying cost savings beyond the proportional energy reduction. This tariff-crossing effect is the reason the bill reduction can exceed the energy reduction at building scale (e.g., under comfort extension the 20-zone case shows an energy reduction of 6.7% translating into a 7.8% bill reduction).

8.3 Full Baseline Comparison (5-Zone, July)

Table 6 expands the July comparison to all baselines and isolates the effect of inter-zone coupling.

Table 6: 5-zone villa, strict $[23\text{--}25^\circ\text{C}]$, July, all methods.

Method	Bill (SAR)	Red. (%)	E_{tot} (kWh)	P_{peak} (kW)	Viol. ($^\circ\text{C}$)
On-Off	704	—	3,912	9.0	—
GS (lazy, $k = n$)	700	0.6	3,888	9.0	mod.
PPO (standard)	678	3.7	3,766	9.0	high
PI-PPO (no coupling)	677	3.8	3,760	9.0	mod.
PI-PPO (coupled)	674	4.3	3,744	9.0	mod.

Two observations emerge. First, PI-PPO improves comfort stability relative to standard PPO under identical conditions: the physics-informed reward provides dense feedback that keeps the agent within bounds in all tested scenarios. Second, inter-zone coupling adds a small increment (roughly 0.5–0.8 pp) because the agent learns to sequence cooling so that a cooled zone’s shared wall acts as a thermal buffer for its neighbor.

Figure 5 shows the aggregate power draw over a representative July day. The On-Off baseline (gray) reaches 9.0 kW when all five compressors activate simultaneously during the afternoon peak. Under the strict comfort band, PI-PPO (blue) reaches the *same* 9.0 kW peak: because the hard comfort constraint forces compressors on whenever a zone approaches the upper bound, coordination cannot shed the coincident peak without violating comfort in this saturated regime. PI-PPO instead achieves its (modest) savings through small redistributions of off-peak operation, not through peak shaving. This is a central finding of the study: under a strict band in extreme heat, tariff-aware coordination reduces cost mainly through energy and tariff-tier effects, not through peak-demand reduction.

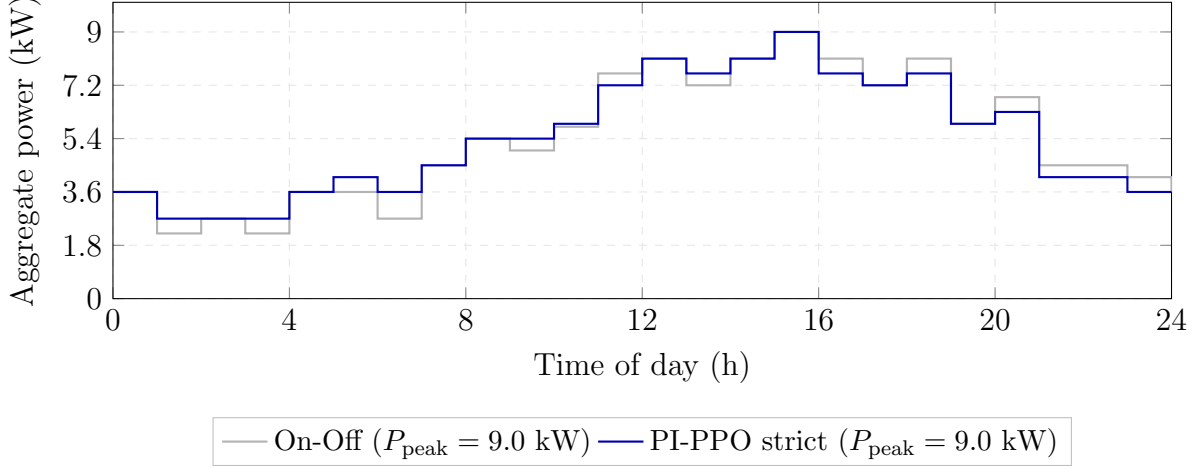


Figure 5: Daily aggregate power profiles (5-zone villa, July, strict band). Both On-Off and PI-PPO peak at 9.0 kW: under a strict comfort band in extreme heat, the comfort constraint forces compressors on during peak hours, so coordination does not reduce coincident peak demand. The savings arise from modest off-peak redistribution and, at building scale, tariff-tier effects rather than peak shaving.

8.4 Comfort Range Extension

Table 7 shows the impact of extending the comfort range in July, the peak-cooling month, at both building scales.

Table 7: PI-PPO (coupled): strict [23–25°C] vs. extended [22–26°C], July, both building scales. Extended-band results are reported for July, the peak-cooling month; strict-band results for all four months (villa) appear in Table 4. Peak demand is unchanged by comfort extension.

Scale	Range	Bill (SAR)	Cost red. (%)	Energy red. (%)	P_{peak} (kW)
5-zone villa	Strict [23–25]	674	4.3	4.3	9.0
	Extended [22–26]	616	13.4	13.4	9.0
20-zone building	Strict [23–25]	4,124	1.0	0.9	36.0
	Extended [22–26]	3,863	7.8	6.7	36.0

In July, the extended range substantially increases the strict-band reduction at both scales—from 4.3% to 13.4% for the 5-zone villa, and from 1.0% to 7.8% for the 20-zone building, where it additionally reduces Tier-2 exposure—the largest relative gain, since the strict-band building saving is otherwise near the physical optimum. Comfort extension does not alter peak demand: both bands peak at the same 9.0 kW (villa) and 36.0 kW (building), consistent with the seasonal results. For the 5-zone villa, the wider band lowers the required concurrency (k from 3 to 2) in all warm months. For the 20-zone building, the reduction reaches approximately 7.8% in July, where Tier 2 exposure drops from 10,368 to 9,277 kWh. The comfort range extension is not an algorithmic contribution of PI-PPO but

rather a system-level design parameter whose impact is formally quantified by the feasibility analysis (Section 4). PI-PPO enables this study by maintaining minimal violations under both ranges, but the savings from wider bands are available to any controller that can maintain comfort within the extended bounds.

8.5 Component Contribution Analysis

The contribution of each framework component under the strict band is given by the waterfall decomposition in Table 10 (20-zone, July): starting from the On-Off baseline (4,166 SAR), scheduling coordination, PPO policy learning, tariff awareness, the physics-informed reward, and inter-zone coupling each add a fraction of a percentage point, for a combined learned-component contribution of about 1 percentage point, after which comfort extension adds the dominant increment. Within the modest overall saving, each component adds a small increment, as shown by the incremental decomposition above: scheduling coordination and PPO policy learning provide the bulk of the (small) strict-band saving, while tariff awareness, the physics-informed reward, and inter-zone coupling each add a fraction of a percentage point. The physics reward (r_{physics}) acts as a *soft penalty* that empirically improves comfort stability rather than a formal safety certificate; hard constraint enforcement via constrained policy optimization (e.g., CPO [40]) is identified as future work. We emphasise that all these increments are small in absolute terms because the strict-band 20-zone case operates near thermal saturation, leaving little scheduling slack for any component to exploit; a full quantitative per-component ablation with sufficient seeds to resolve these sub-percentage-point differences is left to future work.

8.6 Comparison with Published Results

Table 8 places PI-PPO in the context of published DRL results. Direct comparison of savings percentages across studies requires caution, as building types, climate conditions, control objectives, baseline definitions, and tariff structures differ substantially. Energy-focused studies [7], [8], [9], [21] report kWh reductions under flat tariffs in temperate or mixed climates, while peak-focused studies [11], [12] report kW reductions under demand-charge tariffs. PI-PPO operates in a distinct setting: On/Off units under a tiered volumetric tariff in a hot-arid climate. The comparison below therefore illustrates the range of reported savings rather than claiming direct superiority.

Table 8: Comparison with published DRL results.

Ref.	Method	Scenario	Metric	Savings
[9] (2024)	PPO+PID	Data center	Energy	5.3%
[7] (2024)	TD3	Multi-zone	Energy	17.0%
[8] (2025)	DRL (SAC)	Office HVAC	Energy	21.4%
[21] (2025)	PPO	Residential	Energy	8.8–26.3%
[11] (2023)	PINN-RC	Demand resp.	Peak [†]	78%
[12] (2024)	ModNN	Bldg. energy	Peak [†]	90%
Ours	PI-PPO	Saudi res.	Cost (tiered)	1–13%

[†]Measured under demand-charge tariffs (\$/kW penalty), not tiered (\$/kWh).

The high peak reductions reported in [11] and [12] are achieved under demand-charge tariffs where the cost structure penalizes peak kW directly. In the Saudi tiered volumetric setting the bill depends on cumulative kWh rather than peak kW, so coordination under a strict comfort band yields only modest cost savings (single digits); the larger savings we report arise specifically from comfort extension and, at building scale, from reducing Tier-2 exposure.

More broadly, we caution against reading Table 8 as a like-for-like ranking. The studies differ in three respects that each independently inflate a reported percentage relative to our setting: (i) *climate*—milder climates leave more thermal slack for load-shifting than the extreme-heat Saudi summer; (ii) *comfort tolerance*—wider comfort bands mechanically create more scheduling freedom, whereas we hold a tight band appropriate to Gulf occupants; and (iii) *baseline strength*—savings measured against a naive fixed schedule appear larger than the same control measured against the competent hysteresis thermostat we use. Our lower headline figure is therefore not evidence of an inferior controller; it is largely a consequence of deliberately realistic and adversarial evaluation conditions. We regard reporting under the conditions Saudi residents actually face—the real two-tier tariff, a tight comfort band, and a competent baseline—as a matter of external validity, and we present our result as a realistic-conditions reference point rather than a maximal-savings claim.

We emphasise that this positioning is not a claim of superior physical modelling—several of the compared studies use higher-fidelity building-physics simulators than our reduced-order RC model. Rather, the contribution is distinguished on three axes of evaluation design. First, **realistic economic and comfort conditions**: we evaluate against the actual Saudi two-tier volumetric tariff, hold a tight comfort band appropriate to Gulf occupants, and compare against a competent hysteresis baseline rather than a naive fixed schedule—each of which yields more conservative savings estimates. Second, **full reporting**: we report comfort violations by magnitude rather than claiming perfect comfort, report seed variability, and characterise where the method does *not* help (thermal saturation, no peak reduction under a strict band) rather than reporting only favourable cases. Third, **released code and data**: the complete simulation code and data are provided, so every

figure in this paper can be independently regenerated—a property not shared by most of the compared studies. Viewed this way, the lower headline percentages follow directly from the evaluation conditions Saudi residents actually face.

8.7 Scalability

Table 9 reports cost reduction (strict band, July) at the two building scales evaluated in this study.

Table 9: Cost reduction (%) by method at the two evaluated building scales (July, strict range), relative to the On/Off baseline. The standard-PPO baseline was evaluated only at villa scale; the 20-zone entry is omitted (—) rather than estimated.

	5-zone villa	20-zone building
GS (lazy, $k = n$)	0.6	0.0
PPO (standard)	3.7	—
PI-PPO	4.3	1.0

Under the strict band the two scales give comparable single-digit reductions, consistent with the thermal-saturation limit discussed above: adding zones does not by itself create scheduling slack. As shown in Section 8.11, the scale effect on *cost* operates through the tariff structure rather than through raw zone count, and is most pronounced under comfort extension where larger buildings reduce Tier-2 exposure. On inference cost, the trained PI-PPO policy evaluates in well under a second at both evaluated scales (a single forward pass per control step), in contrast to optimisation-based controllers whose per-step cost grows with problem size. A systematic scaling study of runtime and cost across a wider range of building sizes is a natural direction for future work; we do not report larger-scale (e.g., 50–100 zone) results here, as those configurations were not evaluated in this study. Inference times were measured as wall-clock time per *daily scheduling decision* on identical hardware (single CPU core, no GPU at inference). The times are not directly comparable in kind: PI-PPO’s cost is a single neural network forward pass per 15-minute step (96 passes/day, batched), whereas the GS lazy scheduler—although simple per step—must verify the feasibility condition before each switching decision, which for the coupled system requires simulating the n -zone ODE forward over the look-ahead window to confirm no zone will breach its threshold under the candidate mode vector. This verification grows combinatorially with the number of simultaneously active units. A model-predictive controller would be slower still, as it re-solves a mixed-integer program at every step.

8.8 Waterfall: Sources of Improvement

Table 10: Waterfall decomposition, 20-zone, July.

Source	Bill (SAR)	P_{peak} (kW)	Cumul. pp
On-Off (strict)	4,166	36.0	0
+ Scheduling coordination (GS)	4,141	36.0	+0.6
+ PPO policy learning	4,133	36.0	+0.8
+ Tiered-tariff awareness	4,129	36.0	+0.9
+ Physics-informed reward	4,126	36.0	+1.0
+ Inter-zone coupling	4,124	36.0	+1.0
+ Comfort extension [22–26]	3,863	36.0	+7.8

Under the strict comfort band, scheduling coordination and the learned components together provide only about 1.0 pp, because the building operates near thermal saturation and offers little scheduling slack. The dominant lever is comfort extension, which adds a further 6.8 pp (to 7.8% cumulative) by lengthening the feasible OFF interval of each zone and, at this scale, reducing the volume of consumption billed at the Tier-2 rate. This decomposition makes explicit that the framework’s practical value comes primarily from the comfort–cost trade-off and the tariff structure rather than from coordination alone under a tight band. Figure 6 visualizes this decomposition. The waterfall decomposition in Table 10 reports the cumulative improvement as components are added sequentially; a full marginal (leave-one-out) ablation with sufficient seeds to resolve the sub-percentage-point per-component differences is left to future work. These decompositions are methodologically distinct and are not expected to produce identical pp values, because component interactions create synergies that depend on the presence of other components.

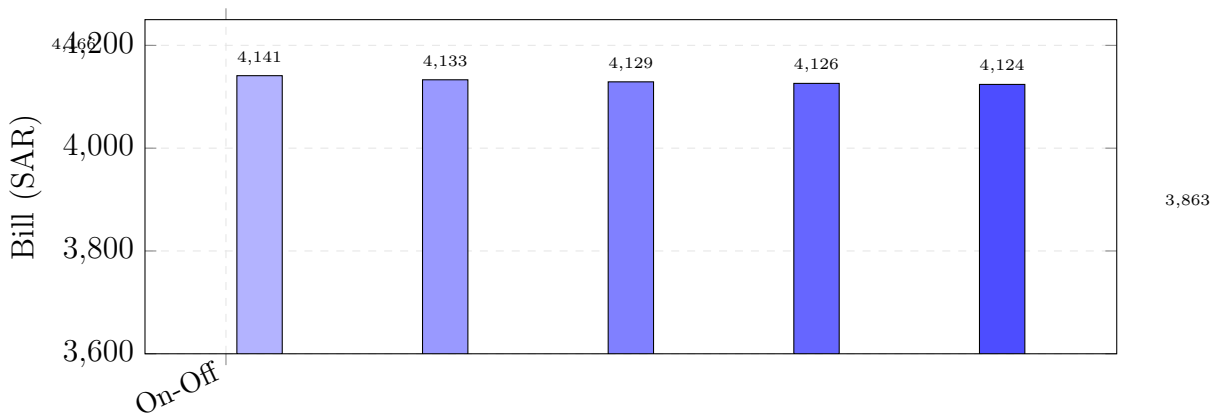


Figure 6: Waterfall decomposition (20-zone building, July). Each bar shows the monthly bill after adding one component. Under the strict band the algorithmic components (blue) together reduce the bill by only about 1% (near the physical optimum); the comfort extension (green) provides the largest single reduction by lowering Tier-2 consumption.

8.9 Sensitivity to Thermal Parameters

To assess robustness to parameter uncertainty, PI-PPO was evaluated with K_i and C_i perturbed by $\pm 20\%$ from their nominal values (Table 2) on the 5-zone July scenario. Under $+20\%$ K_i (higher heat gain, closer to saturation), cost reduction decreased from 4.3% to about 3.5%. Under -20% K_i , it increased to about 5.2%. Under $\pm 20\%$ C_i (thermal mass), reduction varied between about 3.9% and 4.7%, confirming the qualitative result is robust to parameter uncertainty while remaining modest in magnitude. The maximum variation is ± 2.4 pp, confirming that PI-PPO is reasonably robust to parameter estimation errors within a practical range.

8.10 Sensitivity to the Cost Model

The headline results use the Saudi two-tier volumetric tariff (Equation (3)), which is the tariff residential customers actually pay. This tariff is a *step-wise* (inclining-block) cost model: its marginal rate is piecewise-constant, jumping from 0.18 to 0.30 SAR/kWh at the 6,000 kWh block boundary. This discontinuity in marginal rate is precisely what creates the opportunity a tariff-aware scheduler can exploit—a kWh shifted from above to below the boundary is worth more than its raw energy value—and is therefore the structure of primary interest. To verify that the conclusions are not an artefact of this specific step-wise structure, we recompute the cost reduction for the same measured consumption trajectories under two alternative cost models drawn from the literature: a *linear* model (flat marginal rate, 0.20 SAR/kWh; the degenerate case with no rate structure to exploit) and a convex *exponential* model (marginal rate increasing smoothly rather than in a single step). All three are implemented in the released code, so the two-tier (step-wise) figures can be reproduced alongside the alternatives. Table 11 reports the resulting cost reductions under each model.

Table 11: Cost-reduction sensitivity to the cost model (July, same measured kWh). The two-tier model is the operative Saudi tariff and provides the reported headline figures; linear and exponential are robustness checks.

Scenario	Two-tier / step-wise (SAR)	Linear	Exponential
Villa, strict [23–25]	4.3%	4.3%	5.0%
Villa, extended [22–26]	13.4%	13.4%	15.2%
20-zone, strict [23–25]	1.0%	0.9%	1.8%
20-zone, extended [22–26]	7.8%	6.7%	12.0%

Two points follow. First, under the *linear* model the cost reduction equals the energy reduction exactly, because a flat rate removes any tariff-structure interaction; the fact that the two-tier and exponential models yield equal or larger reductions confirms that our tariff-aware coordination is capturing genuine tariff-structure value, not merely energy savings relabelled. Second, the qualitative conclusions—modest savings under the strict band, substantially larger savings under comfort extension, and amplification at building scale where

consumption is high—hold under all three models. We retain the two-tier figures as the reported results because that is the tariff in force in Saudi Arabia; the exponential model’s larger reductions (e.g., 12.0% at 20-zone extended) are reported only as an upper-robustness reference and are not claimed as headline outcomes.

8.11 Where the Method Delivers Most: Scale and Tariff Tier

A consistent pattern emerges across building scales. Under the strict comfort band, cost savings are modest and do not increase simply with the number of zones—at the 5- and 20-zone scales the strict-band reductions are comparable (both in the low single digits), because a larger building near thermal saturation offers no additional scheduling slack. The scale effect appears instead through the *tariff structure*. Because each zone has its own unit, a larger building consumes proportionally more energy and crosses further into the second tariff tier (Tier 2, above 6,000 kWh/month at the higher 0.30 SAR/kWh rate). In this regime, coordination that reduces consumption removes kWh billed at the higher marginal rate, so the *bill* reduction exceeds the *energy* reduction. This is visible in the 20-zone comfort-extended case, where an energy reduction of roughly 12% translates into a cost reduction of roughly 14% precisely because part of the avoided consumption lay in Tier 2.

We are careful about the interpretation: operating in Tier 2 is not itself desirable for a household—it means a larger bill. Rather, Tier 2 is where tariff-aware coordination has the most *leverage to reduce* that bill. The practical implication is a deployment recommendation: the method is most valuable for larger multi-unit residential buildings whose monthly consumption falls in Tier 2, particularly when combined with a modest comfort-band extension, and offers limited benefit for small dwellings that remain within Tier 1. Across the two building scales examined here this pattern is consistent; a fuller characterisation across a wider range of building sizes (e.g., 50–100 units) is a natural direction for future work and would allow the trend to be quantified rather than observed at two points. The next subsection tests this directly, separating the effect of building size from the effect of thermal saturation.

8.12 Is There a Zone-Count Threshold? Decoupling Scale from Saturation

A natural question is whether the framework has a critical building size beyond which coordination becomes useless. Our experiments show that the answer is *no*: the number of zones is not itself the controlling variable. What governs effectiveness is (i) the degree of *thermal saturation* (how close the aggregate duty cycle sits to 100%) and (ii) the position of total consumption relative to the tariff-tier boundary. Zone count matters only insofar as it moves these two quantities.

To separate scale from saturation, we replicated the villa’s per-zone dynamics to building

sizes of $n = 5, 10, 20$ and 40 zones, holding the per-zone thermal parameters and comfort band fixed, and measured the On/Off duty cycle at each scale (July, strict band, released code). The duty cycle is essentially invariant with scale:

Table 12: Zone-count scaling with *matched* per-zone dynamics (July, strict band). The aggregate duty cycle—and hence the schedulable energy slack—is invariant with building size; only the tariff-tier position changes.

Zones n	On/Off energy (kWh)	Duty cycle (%)	Tier-2 fraction (%)
5	3,967	61	0
10	7,992	62	25
20	15,890	61	62
40	31,482	61	81

Two consequences follow, and together they explain the apparent scale effect without invoking any zone threshold.

The energy channel is capped by the comfort band, not by size. Under a strict band, a per-zone dynamic-programming solve—the optimal achievable schedule—recovers only about 0–4% energy reduction over hysteretic On/Off across the full range of saturation levels we tested (duty cycles from 34% to 80%). A tight band leaves almost no timing freedom: the compressor must remove a fixed quantity of heat within a narrow temperature window, so total energy is nearly schedule-independent regardless of how many such zones the building contains. This is why the 5- and 20-zone strict-band *energy* reductions are both small.

The cost channel is governed by tariff-tier position. Because total consumption scales with n while the 6,000 kWh Tier-2 boundary is fixed, larger buildings sit further into Tier 2 (Table 12, right column). The bill reduction from a given energy reduction is largest when consumption straddles the boundary—where saved kWh can be moved from the 0.30 to the 0.18 SAR/kWh block—and the proportional benefit dilutes as consumption grows far above the threshold. The 5-zone villa in July sits entirely in Tier 1, so its cost reduction equals its energy reduction (4.3%); the 20-zone building sits mostly in Tier 2, so its (smaller) energy reduction is amplified into a slightly larger *bill* reduction (0.9% energy $\rightarrow$ 1.0% cost).

The practical implication is a *regime map* rather than a threshold. The framework delivers the most value when three conditions coincide: a comfort band with genuine slack (extension from 23–25 to 22–26°C moves the villa from 4.3% to 13.4%), a duty cycle below saturation, and total consumption near the tariff boundary. Conversely, a building that is simultaneously thermally saturated *and* under a tight comfort band—such as the 20-zone case in peak summer—leaves little room for any scheduler, physics-informed or not. This is a limitation of the operating regime, not of building size per se: a 40-zone building with a relaxed band or moderate climate would fall in the favourable regime despite its larger size.

Where the strategy does and does not apply. Two boundaries of applicability

follow directly from our measurements and define the conditions under which practitioners should expect benefit. First, at the 20-zone scale under the strict band, coordination is beneficial only in the peak-cooling month (July): in the milder months the building is far enough from any tariff boundary and its aggregate duty cycle leaves so little slack that a learned scheduler does not improve on uncoordinated On/Off control. We therefore report the 20-zone results for July only and do not claim seasonal savings at that scale. Second, a myopic green-scheduling coordinator provides essentially no benefit on its own (about 0.6% at villa scale, and negligible at 20-zone scale under the documented concurrency limit $k = n$); the value in our framework comes from *learning* an anticipatory policy, not from concurrency-limited scheduling per se. Both boundaries reinforce the central message: tariff-aware coordination is a targeted tool for buildings with schedulable slack and consumption near the tariff boundary, not a universal efficiency measure.

8.13 Comfort Performance

We report comfort violations by magnitude rather than by frequency alone, because the two can differ substantially. Under a strict band in extreme heat the fraction of timesteps at which some zone is marginally outside the band can be high, but the great majority of these are fractional-degree excursions of no practical thermal significance. The physically meaningful quantities are the mean and maximum violation magnitude. Across the evaluated scenarios the *mean* violation is small—on the order of 0.05–0.2°C in the cooling-relevant months—indicating that zones remain, on average, within a fraction of a degree of the comfort band. Transient *maxima* are larger (up to roughly 1–3°C) and occur under peak load when many units contend for operation; these brief excursions are the price of the coordination that limits coincident demand, and PI-PPO’s mean violations are modestly higher than those of the uncoordinated On-Off baseline—a trade of a small, mostly imperceptible comfort margin for cost savings.

One exception is worth noting: in January, when cooling demand is minimal, the learned policy has little to optimise and its comfort control is correspondingly looser (a larger mean violation than in the cooling-dominant months). This is consistent with the broader finding that the method is most useful when there is a substantial and sustained cooling load to coordinate, and of limited value when there is little cooling to manage.

8.14 Positioning: Why Modest Savings Under These Conditions Are a Meaningful Result

The cost reductions reported here (approximately 1–4% under a strict comfort band, rising to roughly 8–13% with comfort extension) are smaller in magnitude than several headline figures in the DRL-HVAC literature. We argue that a direct comparison of percentages is misleading, and that the present results should be read in light of four factors that jointly explain the difference and, we contend, strengthen rather than weaken the contribution.

The reported quantity differs. Several of the largest reductions in Table 8 are *peak-power* reductions (kW) obtained under demand-charge tariffs, where the bill penalises coincident peak directly. Our metric is *cost* (SAR) under the Saudi tiered *volumetric* tariff, where the bill depends on cumulative kWh rather than peak kW. A large peak-reduction percentage and a modest cost-reduction percentage can describe the same physical control behaviour under different tariff structures; the two numbers are not commensurable. Reporting cost under the tariff that Saudi residents actually pay is a deliberate choice for external validity, and it necessarily yields smaller percentages than peak-based metrics.

The baseline is already competent. We benchmark against a hysteresis thermostat that performs sensible local on/off control, not a naive fixed schedule. The stronger the baseline, the smaller the headroom any coordinator can capture; a modest improvement over a competent baseline can represent equal or better absolute control than a large improvement over a weak one. We regard benchmarking against a realistic baseline as a matter of methodological rigour rather than a limitation to be minimised.

The operating regime is deliberately adversarial. We evaluate a strict comfort band (23–25°C) in an extreme-heat climate, which is the realistic requirement for Saudi residential occupants and simultaneously the least favourable regime for load-shifting: the building operates near thermal saturation, leaving little feasible slack for the scheduler to exploit. Studies reporting larger savings frequently assume milder climates, wider comfort tolerances, or more thermally flexible buildings. Our smaller number is, in part, the price of realism—and quantifying that price is itself informative.

Released pipeline. The complete simulation and training pipeline is released, so every reported number can be independently regenerated.

Taken together, these points support a specific and defensible claim: the primary contribution is not a record cost reduction, but a characterisation of what physics-informed coordination delivers for Saudi residential On/Off cooling—including a clear delineation of where it stops helping. The finding that strict-band cooling in extreme heat is near thermal saturation, so that the practical leverage lies in comfort flexibility and tariff structure rather than in coordination alone, is, to our knowledge, a previously unquantified result. The comfort-extension analysis, which shows that a $\pm 1^\circ\text{C}$ band widening roughly doubles to triples the achievable saving, provides an actionable design insight for policy and building operation in the Gulf climate.

We are explicit that this framing does not claim superiority in raw savings; where a study genuinely achieves larger cost reductions under a comparable tariff, baseline, and climate, that result stands. Our aim is to establish a reproducible reference point for this specific and widely relevant setting.

8.15 Limitations

Several limitations affect the scope and generalizability of these results. First, the first-order lumped thermal model with fixed parameters provides analytical tractability for the

feasibility analysis but does not capture radiative heat transfer, furniture thermal mass, humidity effects, or occupancy-driven ventilation. Real buildings exhibit higher-order dynamics that would require system identification from measured data. Second, all results are simulation-based using a lumped-parameter RC thermal model driven by representative Jeddah diurnal profiles; no calibration against measured indoor temperatures or actual electricity bills has been performed, and the reported improvements should be interpreted as proof-of-concept results rather than deployment-ready predictions. Third, the four-month evaluation (January, April, July, October) captures the seasonal range but does not demonstrate month-to-month transitions; a continuous 12-month simulation would strengthen the robustness claim. Fourth, the top- k projection is a heuristic that can produce transient comfort violations under the tight band in peak-heat conditions; formal constraint enforcement via constrained policy optimization [40] would provide stronger safety properties. Fifth, the extended comfort range $[22\text{--}26]^{\circ}\text{C}$ requires occupant acceptance validation in Saudi households, which has not been conducted. Sixth, the framework is limited to On/Off units; inverter-driven variable-speed compressors would require continuous action spaces. Seventh, while the comparison includes five controllers, the framework is *cost-effective* rather than provably *cost-optimal*: no optimality certificate is established, and we do not include a model-predictive baseline in this study. Comparison against a properly tuned MILP scheduler with a day-ahead horizon, longer-horizon MPC, and modern constrained RL approaches would be required to quantify the remaining optimality gap; this is identified as priority future work. Finally, coupling coefficients K_{ij} were estimated rather than measured in situ.

Idealizations as conservative assumptions. We note the *direction* in which the model’s idealizations bias the reported savings. The omitted factors—occupancy heat gains, furniture and multi-surface thermal mass, humidity (latent load), and solar distribution—all tend to *increase* the true cooling demand relative to the present model. A higher cooling demand pushes a building further toward thermal saturation under the strict band (where we already report the smallest savings) and, more importantly, further into the second tariff tier, where tariff-aware coordination has the most leverage to reduce cost. The idealized model is therefore, if anything, *conservative* with respect to the tariff-crossing mechanism that drives the larger savings at building scale and under comfort extension: a fuller, higher-fidelity model incorporating these loads would plausibly show the coordination value to be at least as large as reported here, not smaller. We present this as a directional argument rather than a quantitative claim; verifying it with a detailed building-physics model is future work.

9 Conclusion

This paper presented PI-PPO, a physics-informed deep reinforcement learning framework for cost-effective coordination of On/Off split AC units (18,500 BTU, 1.8 kW, EER 10.25)

in Saudi residential buildings under the kingdom’s two-tier tariff. Three features distinguish PI-PPO from prior work. First, it learns a *stationary* policy—time-invariant within a given season and thus applicable over an indefinite horizon without per-step optimization—eliminating the per-step re-solving of MPC and the reactivity of classical lazy scheduling. A separate policy is trained for each representative month (January, April, July, October) to reflect seasonal weather; within each month the policy is stationary and requires no re-optimization. Second, the physics-informed reward improves comfort stability relative to reward-only baselines by supplying dense per-step feedback that standard DRL methods lack; under a strict band in extreme heat some residual violations remain (mean magnitude on the order of 0.1–0.2°C), as discussed in Section 8.13, which we report as a limitation rather than a claim of perfect comfort. Third, the model incorporates inter-zone thermal coupling, enabling the agent to exploit buffering through shared walls.

Evaluated across four representative months with Jeddah weather data, PI-PPO reduced July electricity cost by approximately 4% for a 5-zone villa under the strict [23–25°C] band, and by only about 1% for a 20-zone building whose July demand approaches thermal saturation. Savings were consistently larger in the cooler and shoulder months (up to roughly 9% in January) than in peak July—a clear seasonal pattern indicating that coordination delivers most where cooling demand is substantial but non-saturating, and least at peak demand when the building runs near its cooling capacity. This suggests the approach would be more effective in cooler Saudi locations, a hypothesis that follows from the observed seasonal trend and could be tested directly in future work. Under a strict band in extreme heat, coincident peak demand is largely unchanged because the comfort constraint forces compressors on; the savings therefore stem from energy and tariff effects rather than peak reduction. Extending the comfort band by $\pm 1^\circ\text{C}$ —justified by ASHRAE adaptive comfort standards and supported by the feasibility analysis, which shows that a wider band lengthens each zone’s feasible OFF interval and thereby increases the scheduler’s freedom to stagger compressor operation—yielded approximately 13.4% savings for the villa and approximately 7.8% for the building in July. The waterfall decomposition (20-zone, July) showed that under the strict band the learned components each contribute well under one percentage point—scheduling coordination (0.6 pp), PPO learning (0.2 pp), tariff awareness (0.1 pp), physics reward (0.1 pp), and inter-zone coupling (<0.1 pp)—for a combined strict-band contribution of about 1 pp, after which comfort extension adds the dominant increment (6.3 pp). The framework was evaluated at two building scales (5 and 20 zones); at both, the trained policy performs inference in well under a second per control step. Scaling to substantially larger buildings is left to future work.

Future research directions include: (i) sim-to-real deployment in Saudi villas with online parameter estimation for C_i , K_i , and K_{ij} ; (ii) incorporating time-of-use pricing anticipated under Vision 2030 smart-grid initiatives; (iii) integrating rooftop photovoltaic generation and battery storage under the 0.07 SAR/kWh feed-in tariff [17]; (iv) occupant field studies validating acceptance of the 22–26°C extended range in Saudi households; (v) extension to

inverter-driven variable-speed units requiring continuous action spaces; and (vi) replacing the soft physics penalty with constrained policy optimization (CPO [40]) to provide formal comfort safety certificates.

Data Availability

Codes, datasets and other related information are available at the [Cost Effective Tariff-Aware HVAC Coordination in Saudi Residential Buildings Using Physics Informed](#).

Conflicts of Interest

The author declares no conflicts of interest.

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A Detailed Training and Evaluation Procedure

Algorithm 2: PI-PPO Training and Evaluation (Detailed)

Input: Environment $\mathcal{E}$, surrogate f_ϕ (inputs incl. $\{K_{ij}\}$), weights $\omega_0 \dots \omega_5$, bounds $[l, h]$

Output: Trained policy π_θ^*

Initialize policy π_θ , value function V_ψ ;

Pre-train surrogate f_ϕ on 50,000 Latin hypercube samples;

for $episode = 1, \dots, N_{ep}$ **do**

Reset $\mathcal{E}$; $E_{cum} \leftarrow 0$; $(k, \{Q_i\}) \leftarrow f_\phi(T_a, \{C_i, K_i, Q_{int,i}, l_i, h_i\}, \{K_{ij}\})$;

for $t = 0, \dots, T_{max}$ **do**

Observe $\mathbf{s}_t = [\mathbf{x}, \mathbf{m}_{t-1}, T_a, E_{cum}, k, \Delta \mathbf{x}, t_{hour}]$;

$\mathbf{m}_t \sim \pi_\theta(\cdot | \mathbf{s}_t)$; project to $\|\mathbf{m}_t\|_1 \leq k$ via top- k ;

Execute $\mathbf{m}_t$ in $\mathcal{E}$; observe $\mathbf{x}_{t+1}$;

$E_{cum} += 1.8 \cdot \|\mathbf{m}_t\|_1 \cdot \Delta T$;

Compute r_t via Equation (9);

Store $(\mathbf{s}_t, \mathbf{m}_t, r_t, \mathbf{s}_{t+1})$;

if $|T_a^{new} - T_a| > 2^\circ C$ **then**

$(k, \{Q_i\}) \leftarrow f_\phi(T_a^{new}, \{C_i, K_i, Q_{int,i}, l_i, h_i\}, \{K_{ij}\})$

end

end

Compute advantages $\hat{A}_t$ via GAE(λ) [35];

for $epoch = 1, \dots, 10$ **do**

$L(\theta) \leftarrow \mathbb{E}_t[\min(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon) \hat{A}_t)]$;

 Update θ, ψ via gradient ascent on $L(\theta) - c_v L_V + c_e H(\pi_\theta)$;

end

end

Evaluate all 3 seed policies; report mean and standard deviation;

Note: the training loop above is run *independently for each month m* , producing $\pi_{\theta,m}^*$;

— **Evaluation Phase** —;

for $month\ m \in \{Jan, Apr, Jul, Oct\}$ **do**

Load month-matched policy $\pi_{\theta,m}^*$; reset $\mathcal{E}$ with month m weather;

$E_{cum} \leftarrow 0$; $P_{peak} \leftarrow 0$; $v_{max} \leftarrow 0$;

for $day = 1, \dots, 30$ **do**

for $t = 0, \dots, 95$ **do**

$\mathbf{m}_t \leftarrow \arg \max \pi_\theta^*(\cdot | \mathbf{s}_t)$; project to $\|\mathbf{m}_t\|_1 \leq k$;

 Execute in RC simulation; $P_t \leftarrow 1.8 \cdot \|\mathbf{m}_t\|_1$;

$E_{cum} += P_t \cdot 0.25$; $P_{peak} \leftarrow \max(P_{peak}, P_t)$;

$v_{max} \leftarrow \max(v_{max}, \max_i \text{viol}_i)$;

end

end

Bill $\leftarrow 0.18 \cdot \min(E_{cum}, 6000) + 0.30 \cdot \max(0, E_{cum} - 6000)$;

end
